PRODUCTION EFFICIENCY & LINE BALANCING STUDY

Textile finishing and packaging line | Excel-based industrial engineering case study

Project type: Simulated academic case study
Focus: Method study, time study, takt time, bottleneck identification, workstation allocation and line balancing

1. Executive summary

This study evaluates a simulated seven-workstation textile finishing and packaging line. Five illustrative time observations were recorded for each of 12 work elements. Average observed times were converted to standard times using a stated 10% allowance. The current allocation combines fabric spreading and cutting at one workstation, creating a bottleneck above the required takt time. The proposed allocation separates these activities and redistributes downstream tasks while respecting the stated precedence relationships.

Measure	Current state	Proposed balance
Takt time	52.5 sec/unit	52.5 sec/unit
Bottleneck / line cycle	68.2 sec/unit	50.6 sec/unit
Calculated shift capacity	369 units	498 units
Line efficiency	64.5%	87.0%
Total idle time per cycle	169.4 sec	46.2 sec
Capacity vs. 480-unit demand	–111 units	■18 units

The proposed balance meets the modelled demand with a calculated capacity margin of 18 units per shift. These are model outputs based on assumptions, not verified factory performance.

2. Scope and assumptions

Parameter	Assumption
Production process	Textile finishing and packaging of one finished unit
Net available shift time	420 minutes
Shift demand	480 finished units
Time-study sample	Five observations per work element
Allowance	10% applied to average observed time
Line configuration	Seven workstations in both states
Data source	Illustrative simulated values; not actual plant data

3. Process sequence and work elements

Fabric roll check → shade/defect marking → spreading and feeding → cutting → edge hemming → label attachment and dimensional inspection → final visual quality check → folding → primary packing → carton sealing → final count and scan. Label attachment and dimensional inspection are treated as parallel activities after hemming; both must be complete before final visual quality check.

ID	Work element	Avg. observed (s)	Standard time (s)
E1	Fabric roll check	18.0	19.8
E2	Shade / defect marking	22.0	24.2
E3	Fabric spreading and feeding	24.0	26.4
E4	Cutting to required dimensions	38.0	41.8
E5	Edge hemming / finishing	46.0	50.6
E6	Label attachment	20.0	22.0
E7	Dimensional inspection	24.0	26.4
E8	Final visual quality check	18.0	19.8
E9	Folding	28.0	30.8
E10	Polybag / primary packing	22.0	24.2
E11	Carton sealing	12.0	13.2
E12	Final count and scan	8.0	8.8

Total observed work content = 280.0 sec/unit. Total standard work content after the 10% allowance = 308.0 sec/unit.

4. Method and calculations

Takt time = net available production time ÷ customer demand = (420 × 60) ÷ 480 = **52.5 sec/unit**.

Standard time = average observed time × (1 + allowance) = average observed time × 1.10.

Line cycle time = highest standard workload among the workstations. **Shift capacity** = floor(net available seconds ÷ line cycle time). **Line efficiency** = total standard work content ÷ (number of workstations × line cycle time).

5. Current-state diagnosis

In the current allocation, spreading/feeding (24 sec observed) and cutting (38 sec observed) are combined at Workstation 2. Their combined standard workload is 68.2 seconds, which exceeds the 52.5-second takt time. The station therefore limits the modelled line to approximately 369 units per shift, below the 480-unit requirement.

Station	Current allocation	Standard load (s)
WS-1	Fabric roll check + shade/defect marking	44.0
WS-2	Spreading/feeding + cutting	68.2
WS-3	Edge hemming	50.6
WS-4	Label attachment + dimensional inspection	48.4
WS-5	Final visual quality check	19.8
WS-6	Folding	30.8
WS-7	Packing + sealing + final scan	46.2

6. Proposed line balance

The proposed arrangement separates spreading and cutting into individual stations. Downstream tasks are grouped only where the combined standard load remains within takt time. The precedence relationship is maintained: both label attachment and dimensional inspection are completed before final visual inspection.

Station	Proposed allocation	Standard load (s)	Idle time (s)

WS-1	Fabric roll check + shade/defect marking	44.0	6.6
WS-2	Fabric spreading and feeding	26.4	24.2
WS-3	Cutting to required dimensions	41.8	8.8
WS-4	Edge hemming / finishing	50.6	0.0
WS-5	Label attachment + dimensional inspection	48.4	2.2
WS-6	Final visual QC + folding	50.6	-0.0
WS-7	Primary packing + carton sealing + final scan	46.2	4.4

The proposed bottleneck is 50.6 seconds, below the 52.5-second takt time. The theoretical capacity is 498 units per shift, giving a modelled margin of 18 units over demand.

7. Recommendations for a real production environment

• Validate the operation sequence and precedence on the actual shop floor.
• Conduct a fresh time study over representative shifts and product variants.
• Check operator walking distance, material presentation, machine constraints, ergonomics and safety before reallocating work.
• Use 5S and visual work instructions to reduce searching and handling losses.
• Pilot the revised allocation, monitor actual output and quality, then repeat the bottleneck study.
• Review the balance whenever product mix, machine speed or customer demand changes.

8. Limitations and responsible interpretation

This is a simplified, simulated single-product line model. It assumes stable demand, adequate material supply, trained operators, available machines, no major breakdowns, no rework and a uniform allowance. The calculated capacity is theoretical and should not be interpreted as guaranteed production. Real implementation requires plant observation, safety review, quality validation and supervisor approval.

9. Interview-ready project explanation

“I developed an Excel-based line balancing study for a simulated textile finishing and packaging process. I created a time-study sheet with five observations per work element, applied a stated allowance to estimate standard times, calculated takt time from shift demand, and identified a spreading-and-cutting bottleneck. I then proposed a revised workstation allocation and compared line cycle time, capacity, efficiency and idle time before and after the change. The case study helped me apply method study and line balancing concepts to a manufacturing process.”

Deliverable note: The accompanying Excel workbook contains the underlying observations, formulas, workstation allocations, KPI comparison and engineering notes. Open it in Microsoft Excel to recalculate formulas.